TABLE II: AVERAGE CONSTRAINT VIOLATION RATES AND MAXIMUM VIOLATION MAGNITUDES FOR $\mathrm{DCAC}^{\mathrm{BASE}}$, $\mathrm{DCAC}^{\mathrm{SS_{INIT}}}$, AND $\mathrm{DCAC}^{\mathrm{SS_{OPT}}}$ ACROSS 1,000 LOAD SAMPLES (USING $\mathrm{DC_{BASE}}$ SETPOINTS)

| Test case | Method | Active Power (p.u.) | | Reactive Power (p.u.) | | Voltage Limit (p.u.) | | Line Limit (%) | |
|---|---|---|---|---|---|---|---|---|---|
| | | % viol. | Max | % viol. | Max | % viol. | Max | % viol. | Max |
| ieee_118 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 1.61 | 0.01 | 5.56 | 2.93 | 0.05 | 0.01 | 0.00 | 0.20 |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** |
| ACTIVSg_200 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** |
| ACTIVSg_500 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 0.00 | 0.00 | 8.03 | 0.06 | 0.00 | 0.00 | 0.00 | 0.00 |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.03 | 0.71 |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** |
| pegase_1354 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 0.46 | 0.02 | 18.46 | 18.28 | 0.00 | 0.00 | 0.00 | 0.00 |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** |
| pegase_2869 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 0.51 | 2.63 | 15.69 | 13.73 | 0.00 | 0.00 | 0.00 | 0.00 |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.00 | 0.00 | 0.00 | 0.00 | 0.21 | 0.02 | 0.00 | 0.00 |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | **0.00** | **0.00** | **0.00** | **0.00** | **0.03** | **0.01** | **0.00** | **0.00** |
| pegase_9241 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 2.31 | 14.66 | 14.74 | 10.71 | 17.23 | 0.07 | 2.31 | 44.73 |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.00 | 0.00 | 0.00 | 0.00 | 3.75 | 0.02 | 0.00 | 0.00 |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | **0.00** | **0.00** | **0.00** | **0.00** | **2.61** | **0.02** | **0.10** | **0.05** |

TABLE III: PERFORMANCE TABLE FOR $\mathrm{DCAC}^{\mathrm{BASE}}$, $\mathrm{DCAC}^{\mathrm{SS_{INIT}}}$, AND $\mathrm{DCAC}^{\mathrm{SS_{OPT}}}$ ACROSS 1,000 SAMPLES ($\mathrm{DC_{BASE}}$ SETPOINTS)

| Test case | Method | Cost Diff. (%) | | MAE* (p.u.) | | Iter. Count | | Solve Time (s) | |
|---|---|---|---|---|---|---|---|---|---|
| | | val. | %Improv. | val. | %Improv. | val. | %Improv. | val. | %Improv. |
| ieee_118 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 1.488 | **+33.8** | 0.025 | **+8.0** | 4.0 | — | 0.073 | — |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.993 | **+0.8** | 0.023 | 0.0 | 5.0 | −8.0 | 0.269 | **+24.2** |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | 0.985 | — | 0.023 | — | 5.4 | — | 0.204 | — |
| ACTIVSg_200 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 0.035 | — | 0.003 | 0.0 | 2.0 | — | 0.065 | — |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.524 | −13.9 | 0.003 | 0.0 | 4.0 | **+25.0** | 0.258 | **+46.5** |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | 0.597 | — | 0.003 | — | 3.0 | — | 0.138 | — |
| ACTIVSg_500 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 1.383 | **+70.1** | 0.010 | 0.0 | 3.0 | — | 0.113 | — |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 2.196 | **+81.2** | 0.010 | 0.0 | 3.4 | **+5.9** | 0.266 | **+15.4** |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | 0.413 | — | 0.010 | — | 3.2 | — | 0.225 | — |
| pegase_1354 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 0.394 | **+88.8** | 0.050 | **+20.0** | 4.0 | — | 0.361 | — |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.333 | **+86.8** | 0.048 | **+16.7** | 4.8 | **+10.4** | 1.102 | **+19.9** |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | 0.044 | — | 0.040 | — | 4.3 | — | 0.883 | — |
| pegase_2869 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 0.458 | **+95.4** | 0.042 | **+28.6** | 4.0 | — | 0.915 | — |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.387 | **+94.6** | 0.041 | **+26.8** | 5.3 | **+17.0** | 2.324 | **+19.8** |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | 0.021 | — | 0.030 | — | 4.4 | — | 1.865 | — |
| pegase_9241 | $\mathrm{DCAC}^{\mathrm{BASE}}$ | 0.631 | **+80.3** | 0.038 | **+34.2** | 3.0 | — | 3.389 | — |
| | $\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | 0.510 | **+75.7** | 0.035 | **+28.6** | 7.1 | **+15.5** | 10.516 | **+25.3** |
| | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ | 0.124 | — | 0.025 | — | 6.0 | — | 7.860 | — |

The % Improv. report values relative to $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ for each metric. $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ uses optimized values for smooth distributed slack and smooth PV/PQ switching. The initialized smooth pipeline, $\mathrm{DCAC}^{\mathrm{SS_{init}}}$, uses initial parameter values: voltage setpoints initialized to 1.0 p.u., with participation factor 1.0 assigned to the slack bus and 0.0 to all other buses.

TABLE IV: TEST LOSS VALUES ACROSS 1,000 SAMPLES COMPARING $\mathrm{DCAC}^{\mathrm{BASE}}$, $\mathrm{DCAC}^{\mathrm{SS_{INIT}}}$ AND $\mathrm{DCAC}^{\mathrm{SS_{OPT}}}$ (USING $\mathrm{DC_{BASE}}$ SETPOINTS)

| Test Case | Evaluation Loss $\mathcal{L}$ | | |
|---|---|---|---|
| | $^{\star}\mathrm{DCAC}^{\mathrm{BASE}}$ | $^{\star}\mathrm{DCAC}^{\mathrm{SS_{init}}}$ | $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ |
| ieee_118 | 0.212 | 0.029 | **0.026** |
| ACTIVSg_200 | 0.113 | 0.004 | **0.002** |
| ACTIVSg_500 | 0.122 | 0.015 | **0.002** |
| pegase_1354 | 0.131 | **0.001** | 0.002 |
| pegase_2869 | 0.145 | 0.041 | **0.031** |
| pegase_9241 | 0.254 | 0.042 | **0.025** |

$^{\star}\mathrm{DCAC}^{\mathrm{BASE}}$ and $^{\star}\mathrm{DCAC}^{\mathrm{SS_{init}}}$ denote unoptimized pipelines; their losses are evaluated using (4) upon ACPF convergence.

smooth pipelines. Compared with $\mathrm{DCAC}^{\mathrm{SS_{init}}}$, the optimized smooth pipeline reduces the evaluation loss on most systems, including ACTIVSg_200, ACTIVSg_500, pegase_2869, and pegase_9241. For example, the loss decreases from 0.042 to 0.025 on pegase_9241 and from 0.041 to 0.031 on pegase_2869. The baseline produces much larger losses across all cases, confirming that the smooth restoration structure substantially improves recovery quality over conventional single-slack ACPF. This is consistent with the findings in [14]. Table III further shows that optimization improves cost consistency, dispatch accuracy, and computational performance on the larger systems. On pegase_1354, $\mathrm{DCAC}^{\mathrm{SS_{opt}}}$ reduces the CD from 0.333% to 0.044% relative to $\mathrm{DCAC}^{\mathrm{SS_{init}}}$, while reducing MAE from 0.048 to 0.040 p.u. On pegase_2869, the cost difference decreases from 0.387% to 0.021% and MAE reduces from 0.041 to 0.030 p.u. On pegase_9241, the cost difference is lowered also, from 0.510% to 0.124% and MAE follows that trend, from 0.035 to 0.025 p.u. The optimized smooth pipeline also lowered average solve time on these large cases. These results show that optimizing the smooth parameters improves both AC feasibility restoration and ACOPF-tracking performance relative to the initialized smooth pipeline, while remaining substantially better than the single-slack baseline.